Assignment-5

Breast Tumor Classification Using Deep Learning

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Course: Deep learning for medical image processing
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1. Introduction

Breast cancer is one of the most common diseases affecting women across the globe. The early diagnosis of this disease plays a critical role in the effective treatment and improvement of the survival rate. Medical imaging techniques are being used to detect breast cancer. However, analyzing the images obtained from ultrasound scans can be a tedious process and relies on the expertise of medical professionals. In this project, a deep learning-based system will be developed to classify breast ultrasound images into benign, malignant, and normal.

2. Dataset Description

The dataset used for this project is the Breast Ultrasound Images Dataset, denoted as BUSI, where images are categorized based on three types: benign, malignant, and normal. After preprocessing the images, the images were split into training, validation, and test datasets. In the training process, the training dataset comprised 306 benign images, 147 malignant images, and 93 normal images. On the other hand, the validation dataset comprised 65 benign images, 32 malignant images, and 20 normal images. Similarly, the test dataset comprised 66 benign images, 31 malignant images, and 20 normal images. During the preprocessing, the images were resized, and various techniques such as flipping, rotation, and brightness were applied to improve the ability of the model to generalize, thereby improving the performance of the classification model.

3. Methodology

Data Preprocessing: Ultrasound images from the Breast Ultrasound Images Dataset (BUSI) were resized to 224 × 224 pixels and augmented using horizontal flipping, rotation, and brightness/contrast adjustments to improve model generalization. The processed images were then converted into tensor format for deep learning training.

Model Architecture: A transfer learning approach was used with the ResNet-101 convolutional neural network, where the pretrained model's final fully connected layer was replaced to classify images into benign, malignant, and normal categories.

Framework: The deep learning model was implemented using the PyTorch framework, which provides efficient tools for neural network training and GPU acceleration.
Training Configuration: The model was trained using the Adam optimizer with a learning rate of 0.00004 and the CrossEntropyLoss function to handle multi-class classification.
Training Performance: The model was trained for 15 epochs, during which the training loss decreased significantly from 54.56 to 15.79, demonstrating that the network successfully learned important features from the ultrasound images.

4. Results and Performance Evaluation

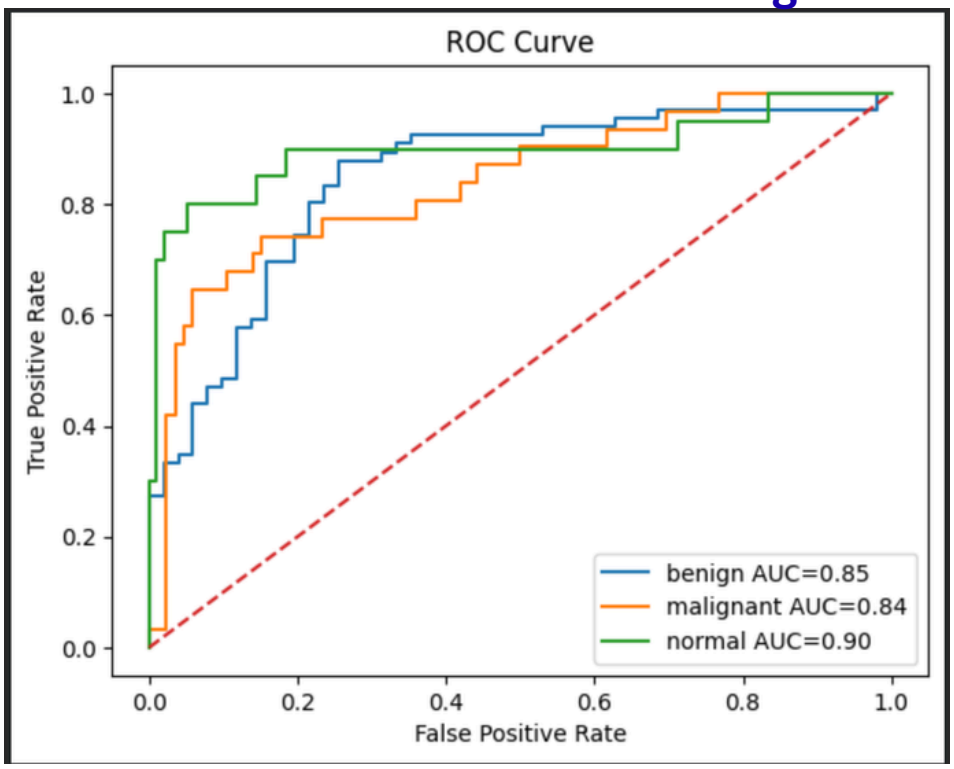
Overall Performance:

Metric	Value
Accuracy	78.6%
Precision	78.5%
Recall	78.6%
F1 Score	78.3%

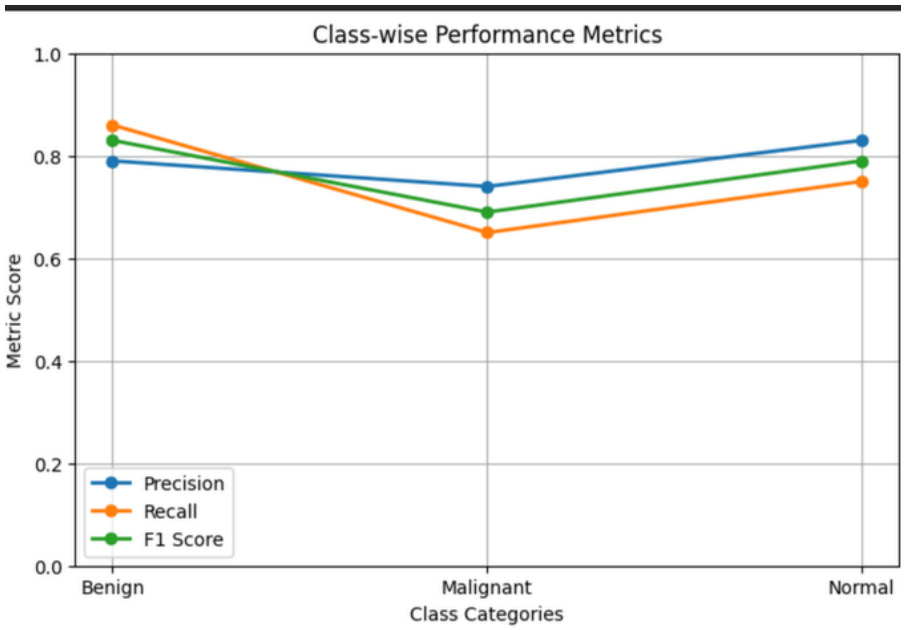
Class-wise Performance:

Class	Precision	Recall	F1 Score	Support
Benign	0.79	0.86	0.83	66
Malignant	0.74	0.65	0.69	31
Normal	0.83	0.75	0.79	20

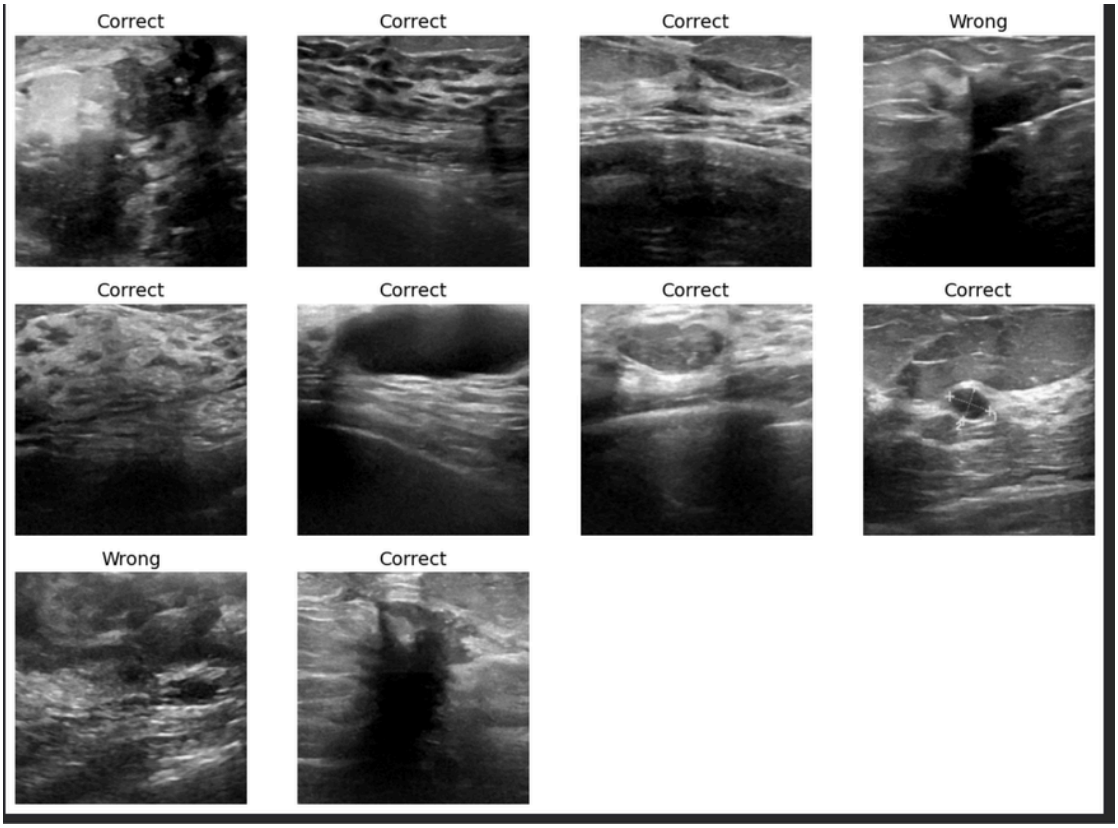
ROC Curve for Breast Ultrasound Image Classification



Class-wise Performance Metrics



Sample Prediction Results on Test Images



Conclusion:

This project proposed an approach for breast tumor classification using ultrasound images with the help of the deep learning technique. An overall accuracy of 78.6% has been obtained using transfer learning with ResNet-101 for distinguishing between benign, malignant, and normal breast images. The results suggest that the proposed deep learning technique can be helpful for medical professionals in the early detection and diagnosis of breast tumors.